

Lecture Notes

EC226 Term 1 2025/26

RHS Endogenous Variables

Readings:

- Stock and Watson (2003): Chapter 12
- Dougherty (2016): Chapters 8 + 9
- Wooldridge (2013): Chapters 15 + 16

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1 RHS Endogeneous variables

Consider the model

$$y_{1i} = \beta_0 + \beta_1 y_{2i} + \beta_2 x_{1i} + \beta_3 x_{2i} + \varepsilon_{1i} \quad (1)$$

where all the CLRM assumptions regarding the error term ε_{1i} are satisfied. However, it may be case that the RHS variable from equation (1), y_{2i} , is an endogenous variable in another equation, in which case the variable y_{2i} is not non-stochastic. In particular, there is an equation in which y_{2i} is the dependent variable, that is,

$$y_{2i} = \delta_0 + \delta_1 z_{1i} + \delta_2 z_{2i} + \delta_3 x_i + \varepsilon_{2i} \quad (2)$$

1.1 Implications of RHS endogenous variables

Suppose it is the case that $cov(\varepsilon_{1i}, \varepsilon_{2i}) > 0$ (< 0), because those unobservable factors that make y_{2i} more likely, $\varepsilon_{2i} > 0$, also make y_{1i} more (less) likely, $\varepsilon_{1i} > 0$ (< 0). For example, consider the situation in which we want to estimate the equation:

$$\ln(W_i) = \beta_0 + \beta_1 S_i + \beta_2 F_i + \beta_3 Age_i + \varepsilon_{1i} \quad (3)$$

Where W - Wages, S - Years of schooling, $F = 1$ if female, 0 otherwise and Age - Age and ε_{1i} is made up of those things, which are unobservable, which affect wages and this might include:

For OLS to be consistent we require that the explanatory variables and the error term are uncorrelated. But schooling (S) is a choice variable, we believe:

$$S_i = f(\dots\dots\dots + \varepsilon_{2i} \quad (4)$$

and ε_{2i} is made up of those things, which are unobservable, which affect schooling and this might include:

The question then arises:

Are those unobservables which mean the individual undertakes more schooling, $\varepsilon_{2i} > 0$, associated with those unobservables which improve wages $\varepsilon_{1i} > 0$, or lower wages, $\varepsilon_{1i} < 0$?

We expect $cov(\varepsilon_{1i}, \varepsilon_{2i}) > 0$, e.g. the unobservables (say ability) which means one's wages are higher, means one also has higher schooling.

Hence

$$\begin{aligned} \text{cov}(S_i, \varepsilon_{1i}) &= \text{cov}(f(\dots) + \varepsilon_{2i}, \varepsilon_{1i}) \\ &= \underbrace{\text{cov}(f(\dots), \varepsilon_{1i})}_0 + \underbrace{\text{cov}(\varepsilon_{2i}, \varepsilon_{1i})}_{\neq 0} \Rightarrow \text{cov}(\text{attend}_i, \varepsilon_{1i}) > 0 \end{aligned}$$

and so OLS estimation of equation (1) yields biased (and inconsistent) estimators. This bias is called endogeneity bias. We have seen other examples in which a bias arises from the use of OLS due to a non-zero correlation between a right hand side explanatory variable and the error term.

1. Measurement error in one of the explanatory variables.
2. An omitted relevant variable.

1.1.1 Simultaneity (as a special case)

Suppose in equation (2) we had instead:

$$y_{2i} = \delta_0 + \delta_1 y_{1i} + \delta_2 z_{1i} + \delta_3 z_{2i} + \delta_4 x_{1i} + \varepsilon_{2i} \quad (5)$$

Now substituting equation (1) into equation (5) and solving for y_{2i} :

$$\begin{aligned} y_{2i} &= \delta_0 + \delta_1(\beta_0 + \beta_1 y_{2i} + \beta_2 x_{1i} + \beta_3 x_{2i} + \varepsilon_{1i}) + \delta_2 z_{1i} + \delta_3 z_{2i} + \delta_4 x_{1i} + \varepsilon_{2i} \\ y_{2i}(1 - \delta_1 \beta_1) &= (\delta_0 + \delta_1 \beta_0) + (\delta_1 \beta_2 + \delta_4) x_{1i} + \delta_1 \beta_3 x_{2i} + \delta_2 z_{1i} + \delta_3 z_{2i} + (\varepsilon_{2i} + \delta_1 \varepsilon_{1i}) \\ y_{2i} &= \pi_0 + \pi_1 x_{1i} + \pi_2 x_{2i} + \pi_3 z_{1i} + \pi_4 z_{2i} + \frac{(\varepsilon_{2i} + \delta_1 \varepsilon_{1i})}{(1 - \delta_1 \beta_1)} \end{aligned} \quad (6)$$

Now for the OLS estimator in equation (1) to be consistent we require that all explanatory variables are unrelated to the error term. However as can clearly be seen in equation (6), $\text{cov}(y_{2i}, \varepsilon_{1i}) \neq 0$; hence OLS will yield biased and inconsistent estimators of the parameters in equation (1).

1.2 IV estimation

To address the problem of the OLS estimator being inconsistent one has to use an alternative estimation method, which essentially breaks the correlation between the explanatory variable and the error term. Instrumental Variable (IV) estimation (or Two Stage Least Squares (TSLS)) is one method econometricians adopt to deal with this correlation.

IV estimation of equation (1) involves running 2 OLS regressions:

1. Regress y_{2i} on the instruments variables outlined in equation (2), (z_{1i}, z_{2i}) and the exogenous variables in equation (1) x_{1i}, x_{2i} and save the fitted values, denoted $\hat{y}_{2i}$.

2. One then estimates the following equation by OLS:

$$y_{1i} = \beta_0 + \beta_1 \hat{y}_{2i} + \beta_2 x_{1i} + \beta_3 x_{2i} + v_{1i} \quad (7)$$

where $v_{1i} = \varepsilon_{1i} + \beta_1(y_{2i} - \hat{y}_{2i})$.

This yields consistent (but not unbiased) estimates of the parameters (see Appendix A for an example of IV estimation).

The IV estimator is written (using the idea of partitioned regression):

$$b_1^{IV} = \frac{\sum_{i=1}^n \tilde{y}_{2i} \tilde{y}_{1i}}{\sum_{i=1}^n \tilde{y}_{2i}^2}$$

and

$$V(b_1^{IV}) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{y}_{2i}^2}$$

where

- $\tilde{y}_{1i}$ are the residuals from a regression of y_{1i} on an intercept x_{1i} , and x_{2i}
- $\tilde{y}_{2i}$ are the residuals from a regression of $\hat{y}_{2i}$ (the predicted values from a regression of the variable y_{2i} on an intercept and x_{1i} , and x_{2i} (1st of the 2-step above)

and this IV estimator is a consistent estimate of β_1 in equation (1) (see Appendix B for some special cases of this IV approach).

In comparison estimating equation (1) by OLS our estimator for β_1 , is written:

$$b_1^{OLS} = \frac{\sum_{i=1}^n \tilde{y}_{2i} \tilde{y}_{1i}}{\sum_{i=1}^n \tilde{y}_{2i}^2}$$

and

$$V(b_1^{OLS}) = \frac{\sigma^2}{\sum_{i=1}^n \tilde{y}_{2i}^2}$$

where

- $\tilde{y}_{2i}$ are the residuals from a regression of y_{2i} on an intercept, x_{1i} and x_{2i}
- $\tilde{y}_{1i}$ are the residuals from a regression of y_{1i} on an intercept, x_{1i} , and x_{2i}

If we compare $V(b_1^{OLS})$ and $V(b_1^{IV})$ as the numerators of both expressions are the same, we want to compare the denominator terms.

- Our OLS estimator $\sum_{i=1} \tilde{y}_{2i}^2$ is the total amount of variation in y_2 (having extracted the influence of x_1 and x_2).
- Our IV estimator $\sum_{i=1} \hat{y}_{2i}^2$ is the amount of variation in $\hat{y}_{2i}$ (the predicted values of y_2 , having extracted the influence of x_1 and x_2).

Given, in general, we can write: $R^2TSS = ESS$ we therefore have:

$$R_1^2 \sum_{i=1} \tilde{y}_{2i}^2 = \sum_{i=1} \hat{y}_{2i}^2$$

where R_1^2 is the R-squared from Step 1 above.

There are 3 interesting cases:

1. If $R_1^2 = 1$, the instruments are perfect in predicting y_2 , then $V(b_1^{OLS}) = V(b_1^{IV})$.
2. If $R_1^2 < 1$, then $\sum_{i=1} \tilde{y}_{2i}^2 > \sum_{i=1} \hat{y}_{2i}^2$ and $V(b_1^{OLS}) < V(b_1^{IV})$ and the weaker the instruments the smaller R_1^2 and the greater the variance of the IV coefficient estimator.
3. If $R_1^2 \Rightarrow 0$ so $\sum_{i=1} \hat{y}_{2i}^2 \Rightarrow 0$ and $V(b_1^{IV}) \Rightarrow \infty$.

In our wage example IV estimation would entail replacing the variable S in equation (3) by a variable which looks like S , but which is unrelated to ε_{1i} , denote this variable as $\hat{S}$. These predicted values could be obtained from regressing S on

Then an OLS estimation of the equation:

$$\ln(w)_i = \beta_0 + \beta_1 \hat{S}_i + \beta_2 F_i + \beta_3 Age_i + \varepsilon_{1i}$$

1.3 Instrument validity

For the instruments we have used to obtain our IV estimator to be good, two conditions MUST satisfied:

1. Instrument relevance – the instruments MUST have a non-zero correlation y_{2i} .
Estimate:

$$y_{2i} = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \delta_3 z_{1i} + \delta_4 z_{2i} + \eta_i$$

and test $H_0 : \delta_3 = \delta_4 = 0$ as an F-test. If the instruments are weak (i.e. they have a low correlation with the endogenous regressions this means the IV results will be inconsistent as:

$$\hat{\beta}_1^{IV} = \beta_1 + \frac{\hat{\beta}_1^{OLS} - \beta_1}{F}$$

where F is the F-test of instrument relevance. Some people have asserted that $F < 10$ means the instruments are weak as this implies a bias of $>10\%$, however, this is clearly just an arbitrary figure.

In our wage example this is a test on the variables in a regression of

$$\hat{S} = f(F, Age, \dots).$$

(see Appendix A for an example of doing this test).

2. Instrument Exogeneity - the instruments must be unrelated to the error term, ε_{1i} in equation (1). This can only be tested if the number of instruments is strictly greater than the number of endogenous regressors. It is tested by as $J = mF \sim \chi^2_{m-k}$ where m is the number of instruments, k is the number of endogenous regressors, and F is an F-test of $H_0 : \gamma_1 = \gamma_2 = 0$ (when $m = 2$) i.e. the joint significance of the instruments in the equation of the IV residuals on both the exogenous variables and the instruments:

$$\hat{u}_i^{IV} = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \gamma_1 z_{1i} + \gamma_2 z_{2i} + v_i$$

where $\hat{u}_i^{IV} = y_{1i} - (\hat{\beta}_0^{IV} + \hat{\beta}_1^{IV} y_{2i} + \hat{\beta}_2^{IV} x_{1i} + \hat{\beta}_3^{IV} x_{2i})$ and are the IV residuals (which are constructed using the true endogenous regressors, NOT the predictions).

In our wage example the regression would be:

$$\hat{u}_i^{IV} = \delta_0 + \delta_1 F_i + \delta_2 Age_i + \gamma_1 \dots + \gamma_2 \dots + \gamma_3 \dots + v_i$$

where $\hat{u}_i^{IV} \equiv \ln(w)_i - (\hat{\beta}_0^{IV} + \hat{\beta}_1^{IV} S_i + \hat{\beta}_2^{IV} F_i + \hat{\beta}_3^{IV} Age_i)$.

(see Appendix A for an example of doing this test).

1.4 Detecting endogeneity: Hausman-Wu test

We want to test the endogeneity of the variable y_{2i} , then you estimate the equation:

$$y_{1i} = \delta_0 + \delta_1 y_{2i} + \delta_2 x_{1i} + \delta_3 x_{2i} + \delta_4 e_{2i} + u_{1i}$$

where e_{2i} are the residual from the instrument relevance test i.e. from a regression of y_{2i} on the exogenous variables and the instruments, by OLS and test $H_0 : \delta_4 = 0$ (i.e. y_{2i} exogenous) vs $H_1 : \delta \neq 0$ (i.e. y_{2i} not exogenous).

This tests works because under H_0 as y_{2i} is exogenous then e_{2i} will contain no information, whereas under H_1 as y_{2i} is endogenous then e_{2i} will contain the important endogenous variation (over above that variation due to the exogenous variables and the instruments) useful for explaining movements in y_{1i} .

1.5 Forbidden Regression

Consider the regression model in which the RHS endogenous variable enters the equation nonlinearly:

$$y_{1i} = \beta_0 + \beta_1 y_{2i} + \beta_1 y_{2i}^2 + \beta_3 x_{1i} + \beta_4 x_{2i} + \varepsilon_{1i} \quad (8)$$

where we have instruments z_{1i} , and z_{2i} . Then one proposed solution to estimating (8) is:

1. Estimate the model:

$$y_{2i} = \delta_0 + \delta_1 z_{1i} + \delta_2 z_{2i} + \delta_3 x_{1i} + \delta_4 x_{2i} + \varepsilon_{2i}$$

and save the fitted values $\hat{y}_{2i}$.

2. Estimate the model:

$$y_{1i} = \beta_0 + \beta_1 \hat{y}_{2i} + \beta_2 \hat{y}_{2i}^2 + \beta_3 x_{1i} + \beta_4 x_{2i} + v_{1i}$$

where $v_{1i} = \varepsilon_{1i} + \beta_1(y_{2i} - \hat{y}_{2i}) + \beta_2(y_{2i}^2 - \hat{y}_{2i}^2)$.

However OLS does not ensure $cov(\hat{y}_{2i}^2, \hat{v}_{1i}) = 0$ as $cov(\hat{y}_{2i}^2, (y_{2i}^2 - \hat{y}_{2i}^2)) \neq 0$ (and this is known as the forbidden regression problem; we see another example of this in Appendix B of the LDV handout).

Instead the approach is:

1. Estimate the model:

$$y_{2i} = \delta_0 + \delta_1 z_{1i} + \delta_2 z_{2i} + \delta_3 z_{1i}^2 + \delta_4 z_{2i}^2 + \delta_5 x_{1i} + \delta_6 x_{2i} + \varepsilon_{2i}$$

by OLS and save the fitted values $\hat{y}_{2i}$.

2. Estimate the model:

$$y_{2i}^2 = \delta_0 + \delta_1 z_{1i} + \delta_2 z_{2i} + \delta_3 z_{1i}^2 + \delta_4 z_{2i}^2 + \delta_5 x_{1i} + \delta_6 x_{2i} + \varepsilon_{3i}$$

by OLS and save the fitted values $\widehat{y_{2i}^2}$.

3. Estimate the model:

$$y_{1i} = \beta_0 + \beta_1 \hat{y}_{2i} + \beta_2 \widehat{y_{2i}^2} + \beta_3 x_{1i} + \beta_4 x_{2i} + v_{1i}$$

where $v_{1i} = \varepsilon_{1i} + \beta_1(y_{2i} - \hat{y}_{2i}) + \beta_2(y_{2i}^2 - \widehat{y_{2i}^2})$ and OLS will ensure $cov(\hat{y}_{2i}, \hat{v}_{1i}) = 0$ and $cov(\widehat{y_{2i}^2}, \hat{v}_{1i}) = 0$ to yield consistent parameter estimates.

Appendices

A 2SLS (or IV) equation estimation

We are wanting to estimate the equation:

$$perf_i = \beta_0 + \beta_1 p_t_abs_i + \beta_2 math_yr1_i + \beta_3 stats_yr1_i + \varepsilon_i$$

where *perf* measures performance in EC226, *p_t_abs* is the proportion of classes missing in EC226, *maths_yr1* is performance in Maths Techniques B and *stats_yr1* is performance in Stats Techniques B. We are using 9am tutorials, fee-status (1=over-seas student, 0 otherwise) and female (1= if female, 0 otherwise) as our instruments for the variable *p_t_abs*.

Estimating by IV yield the same results:

Instrumental variables (2SLS) regression					Number of obs = 149	
					Wald chi2(3) = 128.66	
					Prob > chi2 = 0.0000	
					R-squared = 0.4502	
					Root MSE = 6.0295	

-----+-----						
t2	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
p_t_abs	-35.20629	16.25971	-2.17	0.030	-67.07474	-3.337832
maths	.1522244	.0517821	2.94	0.003	.0507332	.2537155
stats	.2236735	.0602234	3.71	0.000	.1056379	.3417092
_cons	44.51748	4.378827	10.17	0.000	35.93514	53.09983

Instrumented: p_t_abs						
Instruments: maths stats tut 9 female fees_status						

To test for instrument relevance we run the regression:

```
reg p_t_abs tut_9 female fees_status maths stats if(degree=="Econ05" &
yr2_mod=="EC226")
```

Source	SS	df	MS	Number of obs = 149		
Model	.296046069	5	.059209214	F(5, 143)	=	5.80
Residual	1.45942384	143	.010205761	Prob > F	=	0.0001
Total	1.75546991	148	.011861283	R-squared	=	0.1686
				Adj R-squared	=	0.1396
				Root MSE	=	.10102

p_t_abs	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
tut_9	.0713222	.0371033	1.92	0.057	-.0020195	.144664
female	-.0396452	.0183464	-2.16	0.032	-.0759104	-.0033799
fees_status	.0547454	.0199261	2.75	0.007	.0153577	.0941331
maths	.0003168	.0008604	0.37	0.713	-.001384	.0020176
stats	-.0023986	.0007165	-3.35	0.001	-.0038148	-.0009823
_cons	.2309561	.0499526	4.62	0.000	.1322151	.329697

```
test tut_9 female fees_status
```

```
( 1)  tut_9 = 0
( 2)  female = 0
( 3)  fees_status = 0

F( 3, 143) = 4.49
Prob > F = 0.0048
```

The instruments could be said to be weak due to the F-test being <10 , which would potentially make our IV results as biased as the OLS results (in addition to having high standard errors and low t-statistics)

A test for instrument exogeneity yields:

Source	SS	df	MS	Number of obs = 149		
Model	2.77957051	5	.555914103	F(5, 143)	=	0.01
Residual	5414.09717	143	37.8608194	Prob > F	=	0.9999
Total	5416.87674	148	36.6005185	R-squared	=	0.0005
				Adj R-squared	=	-0.0344
				Root MSE	=	6.1531

ivres	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
maths	.002913	.0524074	0.06	0.956	-.1006804	.1065063
stats	-.0020598	.0436389	-0.05	0.962	-.0883205	.0842009
tut_9	.2184255	2.259876	0.10	0.923	-4.248653	4.685504
female	-.1784544	1.117438	-0.16	0.873	-2.387286	2.030377
fees_status	-.2020211	1.213652	-0.17	0.868	-2.601038	2.196996
_cons	.0321451	3.0425	0.01	0.992	-5.98194	6.04623

```
test tut_9 female fees_status
```

```
( 1)  tut_9 = 0
( 2)  female = 0
( 3)  fees_status = 0

F( 3, 143) = 0.02
Prob > F = 0.9948
```

where $J = 3 \times 0.02 = 0.06$, which at the 5% significance level has a critical value of

$\chi^2_2 = 5.99$ and so we are unable to reject the null hypothesis of instrument exogeneity.

B Special case of IV estimation

Consider the special case where we have:

$$y_{1i} = \beta_0 + \beta_1 x_i + \varepsilon_{1i} \quad i = 1, \dots, n \quad (\text{B1})$$

Where $\text{cov}(x, \varepsilon_1) \neq 0$ and we have one instrument z_i :

$$x_i = \delta_0 + \delta_1 z_i + \varepsilon_{2i}$$

and ignoring the idea of deviation from mean we have that our best guesses for x using the instrument z are written:

$$\hat{x}_i = z_i \hat{\delta}_1 = z_i \frac{\sum_{i=1} z_i x_i}{\sum_{i=1} z_i^2}$$

Our IV estimator of β_1 is then written as:

$$\begin{aligned} b^{IV} &= \frac{\sum_{i=1} y_i \hat{x}_i}{\sum_{i=1} \hat{x}_{2i}^2} = \frac{\sum_{i=1} y_i z_i \hat{\delta}_1}{\sum_{i=1} z_i^2 \hat{\delta}_1^2} = \frac{\hat{\delta}_1 \sum_{i=1} y_i z_i}{\hat{\delta}_1^2 \sum_{i=1} z_i^2} = \frac{\sum_{i=1} y_i z_i}{\hat{\delta}_1 \sum_{i=1} z_i^2} \\ &= \frac{\sum_{i=1} y_i z_i}{\left(\frac{\sum_{i=1} z_i x_i}{\sum_{i=1} z_i^2} \right) \sum_{i=1} z_i^2} = \frac{\sum_{i=1} y_i z_i}{\sum_{i=1} z_i x_i} = \frac{\text{cov}(y, z)}{\text{cov}(x, z)} \end{aligned} \quad (\text{B2})$$

Now if $z_i = 1$ $i = 1, \dots, n_1$ and 0 otherwise. Then we note that:

$$\bar{z} = n_1/n = p \text{ and } \bar{y} = p\bar{y}_1 + (1-p)\bar{y}_2, \text{ where } \bar{y}_1 = \sum_{i=1}^{n_1} y_i/n_1, \text{ and } \bar{y}_2 = \sum_{i=n_1+1}^n y_i/n_2.$$

In which case we can write the numerator of b^{IV} as:

$$\begin{aligned} \text{cov}(y, z) &= \frac{\sum_{i=1} y_i z_i - n\bar{z}\bar{y}}{n-1} = \frac{\sum_{i=1}^{n_1} y_i - n\bar{z}\bar{y}}{n-1} \\ &= \frac{n_1\bar{y}_1 - n_1[p\bar{y}_1 + (1-p)\bar{y}_2]}{n-1} = \frac{n_1(1-p)[\bar{y}_1 - \bar{y}_2]}{n-1} \end{aligned}$$

we can then write the denominator of b^{IV} similarly as:

$$\text{cov}(x, z) = \frac{n_1(1-p)[\bar{x}_1 - \bar{x}_2]}{n-1}$$

in which case we have:

$$b^{IV} = \frac{n_1(1-p)(\bar{y}_1 - \bar{y}_2)}{n_1(1-p)(\bar{x}_1 - \bar{x}_2)} = \frac{\bar{y}_1 - \bar{y}_2}{\bar{x}_1 - \bar{x}_2} \quad (\text{B3})$$

C Stata IV commands

```
/* Runnng an IV regression and saving the IV residuals */
ivregress 2sls t2 maths stats (p_t_abs = tut_9 female fees_status) /*
*/ if (degree=="Econ05" & yr2_mod=="EC226")
predict ivres, residuals
/* Performing a 1st astage regression and testing the joint significance of
the instruments */
reg p_t_abs tut_9 female fees_status maths stats if(degree=="Econ05" & yr2_mod=="EC226")
test tut_9 female fees_status
scalar rF=r(F)
/* Perfoming an instrument exogeneity test */
reg ivres maths stats tut_9 female fees_status if(degree=="Econ05" & yr2_mod=="EC226")
test tut_9 female fees_status
scalar Jtest=3*r(F)
```

D R IV commands

#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:

```
library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)
```

```
# Running an IV regression and saving the IV residuals
model26 <- ivreg(t2 ~ p_t_abs + maths + stats | tut_9 female + fees_status,
  data = subset(perf_data, degree=="Econ05" & yr2_mod=="EC226"))
# Performing a 1st a stage regression and testing the joint significance of the instr
model27 <- lm(p_t_abs ~ tut_9 female + fees_status + maths + stats,
  data = subset(perf_data, degree=="Econ05" & yr2_mod=="EC226"))
relev <- linearHypothesis(model27, c("tut_9 female=0", "fees_status=0"), test="F")
t_relev <- relev$F[2]
# Perfoming an instrument exogeneity test
model28 <- lm(resid(model26) ~ maths + stats + tut_9 + female + fees_status,
  data = subset(perf_data, degree=="Econ05" & yr2_mod=="EC226"))
sargan <- linearHypothesis(model28, c("tut_9 female=0", "fees_status=0"), test="F")
Jtest <- 3*sargan$F[2]
```

References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.